MATH 250 — Vector Spaces

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Learning Objectives

- Test sets of vectors for linear independence
- Determine if a set of vectors spans a given space
- Find a basis and dimension for null space and column space
- Apply the Rank-Nullity Theorem and Gram-Schmidt process

For subspace proofs, check all three axioms: zero vector, closure under addition, closure under scalar multiplication.

1. Are u and v linearly independent? Set up cu + dv = 0 and solve.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Answer: _____

2. Determine if the three vectors are linearly independent.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Answer: _____

3. Do u and v span R²? Explain using the determinant.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Answer: _____

4. Is W = {(x, y, z) : x + y + z = 0} a subspace of R³?

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$$

Answer: _____

5. Find a basis and the dimension of the null space of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Answer: _____

6. Find rank(A) and nullity(A). Verify the Rank-Nullity Theorem.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Answer: _____

7. Find a basis for the column space (range) of A.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

Answer: _____

8. Use the Gram-Schmidt process to find an orthogonal basis for the span of v_1 and v_2.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Answer: _____

9. Find the coordinate vector [x]_B of x with respect to basis B.

$$\vec{x} = (5, 3)^T, \quad B = \{(2, 1)^T, (1, 1)^T\}$$

Answer: _____

10. A linear transformation T: $R^2 \to R^2$ is defined by T(x) = Ax. Find $T([2, -1]^T)$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Answer: _____

Problems 4–5: subspace check and null space basis. Problem 6 (Rank-Nullity): critical theorem, always verify n = rank + nullity. Problem 8 (Gram-Schmidt): show the projection formula explicitly.

Solutions

1. Are u and v linearly independent? Set up cu + dv = 0 and solve.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

→ *$c[1,2]^T + d[3,6]^T = 0 \rightarrow c+3d=0,\ 2c+6d=0.$*
→ *Both equations: $c = -3d$. Non-trivial solution exists ($d=1,\ c=-3$).*
→ *$v = 3u$ — the vectors are parallel. Linearly dependent.*

Answer: Linearly dependent: $\vec{v} = 3\vec{u}$

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2. Determine if the three vectors are linearly independent.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

→ *Form matrix $A = [v_1|v_2|v_3]$ and compute det(A).*
→ *$\det([[1,2,1],[0,1,-1],[1,3,0]]) = 1(0+3)-2(0+1)+1(0-1) = 3-2-1 = 0.$*
→ *$\det=0 \rightarrow$ linearly dependent.*
→ *Correction: $1(1\cdot0-(-1)\cdot3)-2(0\cdot0-(-1)\cdot1)+1(0\cdot3-1\cdot1)=1(3)-2(1)+1(-1)=0.$*
→ *Result: $\det = 0 \rightarrow$ vectors are linearly DEPENDENT.*

Answer: Linearly independent

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3. Do u and v span R^2? Explain using the determinant.

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

→ *$\det([[1,-1],[2,3]]) = 3+2 = 5 \neq 0.$*
→ *Non-zero det means the columns are linearly independent.*
→ *{u, v} is a basis for $\mathbf{R}^2$. ✓*

Answer: $\det = 5 \neq 0 \Rightarrow \{\vec{u}, \vec{v}\}$ spans $\mathbb{R}^2$

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4. Is W = {(x, y, z) : x + y + z = 0} a subspace of $\mathbf{R}^3$?

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$$

→ *1) Zero vector: 0+0+0=0 ✓.*
→ *2) Closure under addition: if $x_1+y_1+z_1=0$ and $x_2+y_2+z_2=0$, then $(x_1+x_2)+(y_1+y_2)+(z_1+z_2)=0$ ✓.*
→ *3) Closure under scalar mult: $c(x)+c(y)+c(z)=c(x+y+z)=0$ ✓.*
→ *W satisfies all three subspace axioms.*

Answer: Yes, W is a subspace of $\mathbb{R}^3$

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5. Find a basis and the dimension of the null space of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

→ *Row reduce: already in echelon form. Pivots in cols 1 and 3.*
→ *Free variable: x_2.*
→ *From row 2: $x_3=0$. From row 1: $x_1+2x_2=0 \rightarrow x_1=-2x_2$.*
→ *Null space = {(–2t, t, 0) : t ∈ **R**}. Basis: {[–2,1,0]ᵀ}. dim=1.*

Answer: $\text{Nul}(A) = \text{span}\{(-2,1,0)^T\},\ \dim = 1$

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6. Find rank(A) and nullity(A). Verify the Rank-Nullity Theorem.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ *A is 3×4 (n=4 columns). Already in echelon form.*
→ *Pivot columns: 1 and 3. Rank = 2.*
→ *Nullity = n – rank = 4 – 2 = 2.*
→ *Rank-Nullity: 2+2=4 ✓.*

Answer: $\text{rank} = 2,\ \text{nullity} = 2,\ \text{rank} + \text{nullity} = 4 = n$

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7. Find a basis for the column space (range) of A.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

→ *Row reduce A to find pivot columns.*
→ *R2←R2–2R1, R3←R3–3R1: [[1,2,1],[0,0,1],[0,0,1]].*
→ *R3←R3–R2: [[1,2,1],[0,0,1],[0,0,0]]. Pivots in cols 1 and 3.*
→ *Basis for Col(A) = original pivot columns = [1,2,3]ᵀ and [1,3,4]ᵀ.*

Answer: $\text{Col}(A) = \text{span}\{(1,2,3)^T,\ (1,3,4)^T\}$

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8. Use the Gram-Schmidt process to find an orthogonal basis for the span of v_1 and v_2.

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

→ *$u_1 = v_1 = [1,1,0]^T$.*
→ *$\text{proj}_{u_1}(v_2) = (v_2 \cdot u_1 / u_1 \cdot u_1)u_1 = (1/2)[1,1,0]^T = [1/2,1/2,0]^T$.*
→ *$u_2 = v_2 - \text{proj} = [1,0,1]^T - [1/2,1/2,0]^T = [1/2,-1/2,1]^T$.*
→ *Verify: $u_1 \cdot u_2 = 1/2-1/2+0 = 0$ ✓.*

Answer: $\vec{u}_1 = (1,1,0)^T, \quad \vec{u}_2 = (1/2, -1/2, 1)^T$

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9. Find the coordinate vector [x]_B of x with respect to basis B.

$\vec{x} = (5, 3)^T, \quad B = \{(2, 1)^T, (1, 1)^T\}$

→ *Solve $c_1[2,1]^T + c_2[1,1]^T = [5,3]^T$.*
→ *$2c_1+c_2=5$ and $c_1+c_2=3$.*
→ *Subtract: $c_1=2$. Then $c_2=1$.*
→ *$[x]_B = [2,1]^T$.*

Answer: $[\vec{x}]_B = (2, 1)^T$

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10. A linear transformation T: $\mathbf{R}^2 \to \mathbf{R}^2$ is defined by T(x) = Ax. Find $T([2, -1]^T)$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

→ *$T(x) = Ax = [[1,2],[3,–1]]\cdot[2,–1]^T$.*
→ *Row 1: 1(2)+2(–1) = 0.*
→ *Row 2: 3(2)+(–1)(–1) = 7.*
→ *$T([2,–1]^T) = [0,7]^T$.*

Answer: $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$